

Economia Para Engenharia $\Rightarrow$ Exercícios aula 07

$$1) PMT = \frac{PV \cdot i}{1 - (1+i)^{-n}} \Rightarrow 110.000 = \frac{1.000.000 \cdot i}{1 - (1+i)^{-60}} \Rightarrow 1.000.000 = \frac{1.000.000 \cdot i}{1 - (1+i)^{-60}} \Rightarrow 1.000.000 \cdot (1 - (1+i)^{-60}) = 1.000.000 \cdot i \Rightarrow 1 - (1+i)^{-60} = i \Rightarrow (1+i)^{-60} = 1 - i$$

$$(1,0085)^{-60} = \frac{1}{1,63862} = 0,6103 \Rightarrow 1 - (1+i)^{-n} = 1 - 0,6103 = 0,3897$$

$$\frac{0,0085}{0,3897} \approx 0,02182 \Rightarrow PMT = 1.000.000 \cdot 0,02182 = 2182$$

$$2) PV = PMT \cdot \frac{1 - (1+i)^{-n}}{i} ; (1+i)^6 = 1,06152$$

$$(1+i)^{-6} = \frac{1}{1,06152} \approx 0,94204 \Rightarrow 1 - 0,94204 = 0,05795$$

$$\frac{0,05795}{0,01} = 5,79547 \Rightarrow PV = 1000 \cdot 5,79547 = 5795,47$$

$$3) m = 35 \cdot 12 = 420 \text{ meses}; i_m = (1,0725)^{1/12} - 1 \approx 0,007264 \text{ (0,7264 \% a.m.)}$$

$$FV = 1.000.000 \Rightarrow PMT = \frac{FV \cdot i_m}{(1+i_m)^n - 1} \Rightarrow PMT = \frac{1.000.000 \cdot 0,007264}{2,567 - 1}$$

$$= \frac{1.000.000 \cdot 0,007264}{1,567} \approx 1.445,00 \text{ reais}$$

$$4) 888 \cdot \frac{1 - (1+i)^{-48}}{i} = 31.500 \Rightarrow 1 - (1+i)^{-48} = \frac{31.500 \cdot i}{888} \Rightarrow 1 - (1+i)^{-48} = 35,475 \cdot i$$

$$(1+i)^{48} - 1 = 13,2778 \% \text{ a.a.} \Rightarrow i = 1,0514785 \% \text{ a.m.}$$

$$5) m = 24 \text{ meses}; FV = 30.000; i_m = 0,5654 \% \text{ a.m.}$$

$$PMT = \frac{FV \cdot i_m}{(1+i_m)^n - 1} = \frac{30.000 \cdot 0,005654}{(1,015654)^{24} - 1} = 1149,9 \text{ reais}$$

6) 1) 25% de 25k = $0,25 \cdot 25000 = R\$ 12.500$

Parcelas: $48 \cdot 387,52 = R\$ 17.400,96 \Rightarrow \text{Total} = 12.500 + 17.400,96 = 53.300,96$

2) $0,5 \cdot 50.000 = 25.000$; $24 \cdot 1173,34 = 28.160,16 \Rightarrow \text{Total} = 25k + 28.160 = 53.160,16$

3) 50.000 $\therefore$ À vista é mais vantajoso

7) $A = \frac{PV}{n} = \frac{200.000}{120} = 1666,67$; $J_1 = PV \cdot i = 200.000 \cdot 0,01 = 2000$

$P_1 = A + J_1 = 1666,67 + 2000 = 3666,67$; e vai diminuindo c/ o tempo

$P = PV \cdot \frac{i(1+i)^n}{(1+i)^n - 1} \Rightarrow P = 200.000 \cdot \frac{0,01(1,01)^{120}}{(1,01)^{120} - 1} = 200k \cdot 0,01435 = 2.870$

Juros = $200.000 \cdot 0,01 = 2000$; Amortização = $2870 - 2000 = 870$

8) $PV = 40000$; $n = 6$ meses; $i_{doc} = 2,25\% \text{ a.m.}$; $i_{fixe} = 2,22\% \text{ a.m.}$

Soc: $A = \frac{PV}{n} = \frac{40k}{6} = R\$ 6.666,67$; $J_t = SD_{t-1} \cdot i$; $P_t = A + J_t$; $SD_t = SD_{t-1} - A$

$P = PV \cdot \frac{i(1+i)^n}{(1+i)^n - 1} \Rightarrow P = 40000 \cdot \frac{0,022(1,022)^6}{(1,022)^6 - 1} = R\$ 7134,24$ fixa

SAC é + vantajoso

9) $A = \frac{PV}{n} = \frac{1.000.000}{120} = 8333,33$; $J_1 = PV \cdot i = 10.000$; $P_1 = A + J_1 = 18333,33$

$P_1 - \text{aluguel} = 18.333,33 - 7500 = 10833,33 \Rightarrow FV = PMT \cdot \frac{(1+i)^n - 1}{i}$

$100.000 = 10833 \cdot \frac{(1,01)^m - 1}{0,01} \Rightarrow 1,01^m = 1,923 \Rightarrow m = \frac{\ln(1,923)}{\ln(1,01)} = \frac{0,654}{0,0035} = 65,7$ meses